

Video 3: MECH5200 Numerical Methods for PDEs : *1-Dimensional FD Expressions*

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How to solve a simple PDE using a computer

- We would like to determine, u the deflection of a string under some load f :

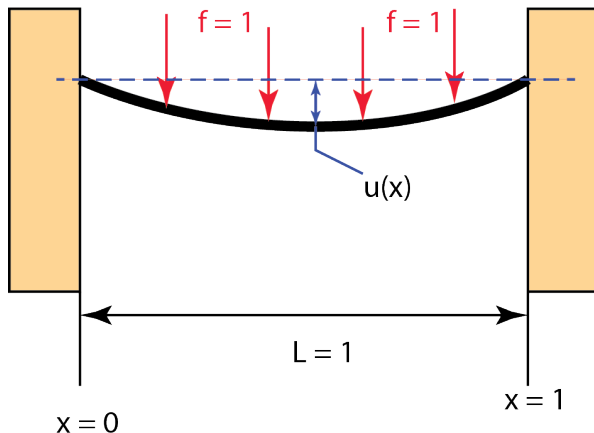


Figure: String Deflection Problem: Finite Difference Example # 1

How to solve a simple PDE using a computer

- Consider the 1-D Poisson Equation (ODE)
- We would like to find the value u between $x = 0$ and $x = 1$:

$$\nabla^2 u = f \quad (1)$$

$$\frac{\partial^2 u}{\partial x^2} = f \quad (2)$$

$$\frac{\partial^2 u}{\partial x^2} = 1 \quad (3)$$

$$(4)$$

- At the ends of the string, the deflection u is zero:

$$u(x = 0) = 0 \quad (5)$$

$$u(x = 1) = 0 \quad (6)$$

The Exact Solution

Determine the exact solution for this problem and bring it to next class!!!

Finite Difference Approximations to Derivatives

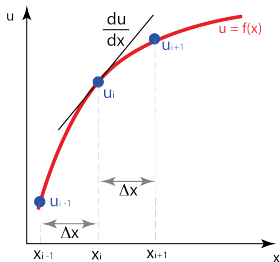
- Let's say we wish to approximate a derivative:

$$\frac{du}{dx} = ??? \quad (7)$$

- How can we numerically approximate this derivative?

Finite Difference Approximations to Derivatives

- Fundamental definition of the derivative:
 - Take the difference of the value for u at two different x - locations and divide by the distance between the points (difference in the x -locations).



$$\frac{du}{dx} = \frac{u_{i+1} - u_i}{x_{i+1} - x_i} + \text{error} \quad (8)$$

Or, we can drop the error term and acknowledge the approximation:

$$\frac{du}{dx} \simeq \frac{u_{i+1} - u_i}{x_{i+1} - x_i} \quad (9)$$

Finite Difference Approximations to Derivatives

- How accurate is this approximation?
- As $\Delta x = (x_{i+1} - x_i) \rightarrow 0$? how rapidly does the numerical approximation of the derivative approach the true derivative

Finite Difference Approximations to Derivatives

- The idea is to approximate derivatives using **finite** differences in the variables.
- To make this a viable method, we need:
 - **GOAL # 1:** Come up with a way to derive derivatives as **finite differences**, eg:
 - $\frac{\partial^2 u}{\partial x^2} = ?$
 - $\frac{\partial^3 u}{\partial x^3} = ?$
 - **GOAL # 2:** Quantify and reduce the error of the **finite difference** approximation of the derivative.
 - **GOAL # 3:** Develop solutions to PDEs using these derivative representations.

Mathematics: Taylor Series Expansion

- **GOAL #1, Method 1: Deriving finite differences**

- The first approach uses a Taylor Series Expansion in positive x-direction.

- Recall the T-S expansion about a point x_0 to determine $u(x_0 + \Delta x)$:

$$u(x_0 + \Delta x) = u(x_0) + \frac{\partial u}{\partial x}|_0 \Delta x + \frac{\partial^2 u}{\partial x^2}|_0 \frac{(\Delta x)^2}{2!} + \dots + \frac{\partial^n u}{\partial x^n}|_0 \frac{(\Delta x)^n}{n!}$$

Geometry and Mathematics of the Taylor Series Expansion

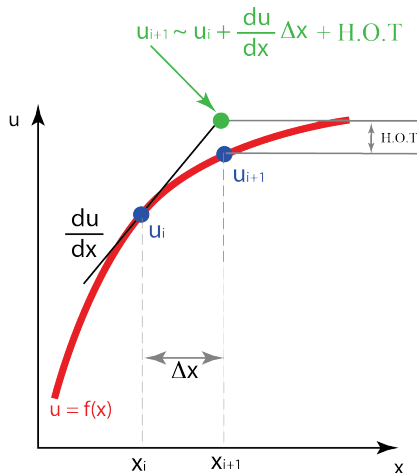


Figure: Taylor Series Expansion – First Term $u_{i+1} \simeq u_i + \left. \frac{\partial u}{\partial x} \right|_i \Delta x + \dots \text{H.O.T}$

Finite Difference Approximations to Derivatives

- The T-S expansion from the previous slide is:

$$u(x_0 + \Delta x) = u(x_0) + \frac{\partial u}{\partial x}\bigg|_0 \Delta x + \frac{\partial^2 u}{\partial x^2}\bigg|_0 \frac{(\Delta x)^2}{2!} + \dots + \frac{\partial^n u}{\partial x^n}\bigg|_0 \frac{(\Delta x)^n}{n!}$$

- We can re-arrange the equation to isolate the first derivative, so that:

$$\frac{\partial u}{\partial x}\bigg|_0 = \frac{(u(x_0 + \Delta x) - u(x_0))}{\Delta x} - \frac{\frac{\partial^2 u}{\partial x^2}\big|_0 \frac{(\Delta x)^2}{2!}}{\Delta x} \dots - \frac{\frac{\partial^n u}{\partial x^n}\big|_0 \frac{(\Delta x)^n}{n!}}{\Delta x}$$

Finite Difference Approximations to Derivatives

- The resulting **forward difference** FD expression for the first derivative is:

$$\frac{\partial u}{\partial x} \Big|_0 = \frac{(u(x_0 + \Delta x) - u(x_0))}{\Delta x} - O(\Delta x) \quad (10)$$

- Comment: The error decays proportionally to decreases in Δx

$$\frac{\partial u}{\partial x} \Big|_i = \frac{(u(x_{i+1}) - u(x_i))}{\Delta x} - O(\Delta x) \quad (11)$$

- Or:

$$\frac{\partial u}{\partial x} \Big|_i = \frac{(u_{i+1} - u_i)}{\Delta x} - O(\Delta x) \quad (12)$$

Finite Difference Approximations to Derivatives

- If we start with:

$$u(x_0 - \Delta x) = u(x_0) - \frac{\partial u}{\partial x} \Big|_0 \Delta x + \frac{\partial^2 u}{\partial x^2} \Big|_0 \frac{(\Delta x)^2}{2!} - \dots + (-1)^n \frac{\partial^n u}{\partial x^n} \Big|_0 \frac{(\Delta x)^n}{n!}$$

- We can re-arrange the equation to isolate the 1st derivative:

$$\frac{\partial u}{\partial x} \Big|_0 = \frac{(u(x_0) - u(x_0 - \Delta x))}{\Delta x} + \frac{\frac{\partial^2 u}{\partial x^2} \Big|_0 \frac{(\Delta x)^2}{2!}}{\Delta x} - \dots + (-1)^n \frac{\frac{\partial^n u}{\partial x^n} \Big|_0 \frac{(\Delta x)^n}{n!}}{\Delta x}$$

Finite Difference Approximations to Derivatives

- The resulting **backward difference** FD expression is:

$$\frac{\partial u}{\partial x} \Big|_0 = \frac{(u(x_0) - u(x_0 - \Delta x))}{\Delta x} + O(\Delta x)$$

- Comment: The error again decays proportionally to decreases in Δx

$$\frac{\partial u}{\partial x} \Big|_i = \frac{(u(x_i) - u(x_{i-1})))}{\Delta x} + O(\Delta x)$$

- Or:

$$\frac{\partial u}{\partial x} \Big|_i = \frac{(u_i - u_{i-1})}{\Delta x} + O(\Delta x)$$

Finite Difference Approximations to Derivatives

- If we start with:

$$u(x_0 + \Delta x) = u(x_0) + \frac{\partial u}{\partial x} \Big|_0 \Delta x + \frac{\partial^2 u}{\partial x^2} \Big|_0 \frac{(\Delta x)^2}{2!} + \dots + \frac{\partial^n u}{\partial x^n} \Big|_0 \frac{(\Delta x)^n}{n!}$$

- And subtract the expression:

$$u(x_0 - \Delta x) = u(x_0) - \frac{\partial u}{\partial x} \Big|_0 \Delta x + \frac{\partial^2 u}{\partial x^2} \Big|_0 \frac{(\Delta x)^2}{2!} - \dots + (-1)^n \frac{\partial^n u}{\partial x^n} \Big|_0 \frac{(\Delta x)^n}{n!}$$

- The result is:

$$\frac{\partial u}{\partial x} \Big|_0 = \frac{u(x_0 + \Delta x) - u(x_0 - \Delta x)}{2\Delta x} + \frac{1}{2} \frac{\partial^3 u}{\partial x^3} \Big|_0 \frac{(\Delta x)^3}{3!} + \dots$$

Finite Difference Approximations to Derivatives

The resulting **central difference** FD expression is:

$$\frac{\partial u}{\partial x} \Big|_0 = \frac{u(x_0 + \Delta x) - u(x_0 - \Delta x)}{2\Delta x} + \frac{1}{2} \frac{\partial^3 u}{\partial x^3} \Big|_0 \frac{(\Delta x)^3}{3!} + \dots$$

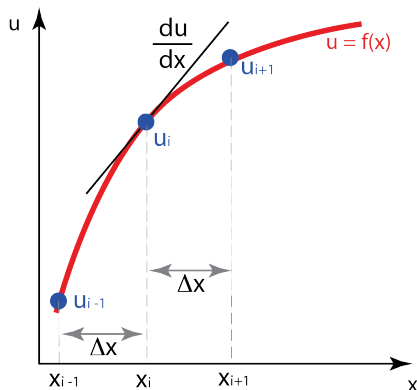
- Which can also be written as:

$$\frac{\partial u}{\partial x} \Big|_i = \frac{u_{i+1} - u_{i-1}}{2\Delta x} + O(\Delta x)^2$$

- This is a more accurate approximation to the first derivative...
- As we reduce the spacing between points by factor of 2, the error in approximating the derivative as a finite difference changes by a factor of 4

Finite Difference Approximations to Derivatives

- Revisiting the original image:



Finite Difference Approximations to Derivatives

- Let's look at a couple of common first derivative finite difference approximations:

$$\frac{du}{dx} = \frac{u_{i+1} - u_i}{\Delta x} + O(\Delta x) \quad (13)$$

$$\frac{du}{dx} = \frac{u_i - u_{i-1}}{\Delta x} + O(\Delta x) \quad (14)$$

$$\frac{du}{dx} = \frac{u_{i+1} - u_{i-1}}{2\Delta x} + O(\Delta x^2) \quad (15)$$

$$\frac{du}{dx} = \frac{3u_i - 4u_{i-1} + u_{i-2}}{2\Delta x} + O(\Delta x^2) \quad (16)$$

$$\frac{du}{dx} = \frac{-3u_i + 4u_{i+1} - u_{i+2}}{2\Delta x} + O(\Delta x^2) \quad (17)$$

Try to derive the second derivative now

- Applying the same approach, see if you can find the second derivative approximation using Taylor series expansions.

$$\frac{d^2 u}{dx^2} = \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2} + O(\Delta x^2)$$

Finite Difference Approximations to Derivatives

- Here are some common finite difference formulas:

$$\frac{d^2 u}{dx^2} = \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2} + O(\Delta x^2)$$

$$\frac{d^3 u}{dx^3} = \frac{u_{i+2} - 2u_{i+1} + 2u_{i-1} - u_{i-2}}{2\Delta x^3} + O(\Delta x^2)$$

$$\frac{d^4 u}{dx^4} = \frac{u_{i+2} - 4u_{i+1} + 6u_i - 4u_{i-1} + u_{i-2}}{\Delta x^4} + O(\Delta x^2)$$

- And a x,y-cross-derivative for good measure:

$$\frac{d^2 u}{dx dy} = \frac{1}{2\Delta x} \left(\frac{u_{i+1,j+1} - u_{i+1,j-1} - u_{i-1,j+1} + u_{i-1,j-1}}{2\Delta y} \right) + O(\Delta x^2, \Delta y^2)$$

- See Tennehill, Anderson, Pletcher, Computational fluid mechanics and heat transfer.
- See also:

web.media.mit.edu/~crtaylor/calculator.html

Finite Difference Equations by the Method of Lagrange Interpolation

- **Method 2:** We can also use **Lagrange Interpolation** to determine the derivatives (and gain insight into FD formulas).

$$\hat{u}(x) = \sum_{j=-left}^{right} L_j(x) u_j \quad (18)$$

- This is a more rigorous and involved approach, but leads to a generalizable method.
- When deriving new finite difference formula, we want to find an approximation of the form:

$$\frac{d^m u}{dx^m} \simeq \sum_{j=-left}^{right} \delta_j^m u_j \quad (19)$$

- We need a way to find the values of the coefficients: δ_j^m

FD Equations by the Method of Lagrange Interpolation

- The Lagrange interpolation is:

$$\hat{u}(x) = \sum_{j=-left}^{right} L_j(x) u_j \quad (20)$$

- Where a general Lagrange Polynomial:

$$L_j(x) = \frac{(x - x_l) \dots (x - x_{j-1})(x - x_{j+1}) \dots (x - x_r)}{(x_j - x_l) \dots (x_j - x_{j-1})(x_j - x_{j+1}) \dots (x_j - x_r)} \quad (21)$$

- The values for the above equation are 1 when $x = x_j$ and 0 when $x_i \neq x_j$.
- Here, x_i is a node other than x_j
- For our example, let's consider finite difference equations that are based on a 3-point stencil (i.e., 3 x Lagrange polynomials that are fit through the 3 points).

FD Equations by the Method of Lagrange Interpolation

- Let's start by approximating the function $u(x)$ (our solution) as the summation of appropriate Lagrange Polynomials:

$$\hat{u}(x) = \sum_{j=-left}^{right} L_j(x) u_j \quad (22)$$

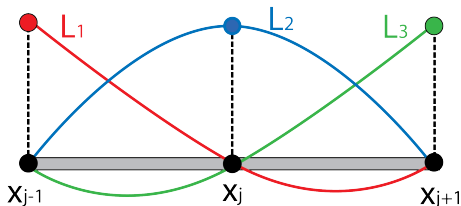


Figure: The three Lagrange polynomials, $L_j(x)$ defined using three x_j values.

FD Equations by the Method of Lagrange Interpolation

- Let's start by approximating the function $u(x)$ (our solution) as the summation of appropriate Lagrange Polynomials:

$$\hat{u}(x) = \sum_{j=-left}^{right} L_j(x) u_j \quad (23)$$

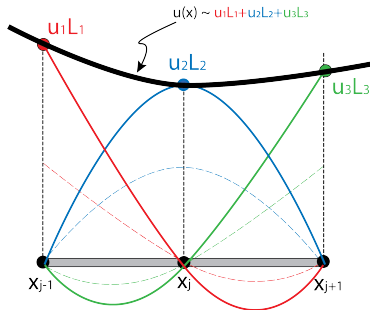


Figure: The Lagrange Polynomial Interpolation scales the Lagrange Polynomials by the nodal u values to produce an (approximate) polynomial interpolation of the function $u(x)$.

FD Equations by the Method of Lagrange Interpolation

- We will examine this approach by example.
- Recall, we want to find an approximation of the form:

$$\frac{d^m u}{dx^m} \simeq \sum_{j=-left}^{right} \delta_j^m u_j \quad (24)$$

- The one-sided first derivative finite difference equation from before:

$$\frac{du}{dx} = \frac{-3u_j + 4u_{j+1} - u_{j+2}}{2\Delta x} \quad (25)$$

- Can be re-written in the form:

$$\frac{du}{dx} = [\delta_j] u_j + [\delta_{j+1}] u_{j+1} + [\delta_{j+2}] u_{j+2} \quad (26)$$

- With the following terms:

$$\frac{du}{dx} = \left[\frac{-3}{2\Delta x} \right] u_j + \left[\frac{4}{2\Delta x} \right] u_{j+1} + \left[-\frac{1}{2\Delta x} \right] u_{j+2} \quad (27)$$

FD Equations by the Method of Lagrange Interpolation

- Start first by expressing the solution as the summation of weighted Lagrange polynomials:

$$\begin{aligned}\hat{u}(x) &= \sum_{j=-l}^r L_j(x) u_j \\ &= \frac{(x - x_j)(x - x_{j+1})}{(x_{j-1} - x_j)(x_{j-1} - x_{j+1})} u_{j-1} + \frac{(x - x_{j-1})(x - x_{j+1})}{(x_j - x_{j-1})(x_j - x_{j+1})} u_j + \frac{(x - x_{j-1})(x - x_j)}{(x_{j+1} - x_{j-1})(x_{j+1} - x_j)} u_{j+1}\end{aligned}$$

- Which (denominator) can be simplified as follows:

$$\begin{aligned}\hat{u}(x) &= \sum_{j=-l}^r L_j(x) u_j \\ &= \frac{(x - x_j)(x - x_{j+1})}{(\Delta x)(2\Delta x)} u_{j-1} - \frac{(x - x_{j-1})(x - x_{j+1})}{(\Delta x)(\Delta x)} u_j + \frac{(x - x_{j-1})(x - x_j)}{(2\Delta x)(\Delta x)} u_{j+1}\end{aligned}$$

FD Equations by the Method of Lagrange Interpolation

- Rather than the solution $\hat{u}$, we want to represent/approximate/focus on the derivative, $\frac{du}{dx}$.

$$\frac{du}{dx} \simeq \sum_{j=-left}^{right} \frac{dL_j}{dx} u_j \quad (28)$$

- What this means, is that the coefficients δ_j^m that we wish to find are simply (by pattern matching):

$$\delta_j = \frac{dL_j}{dx} \quad (29)$$

FD Equations by the Method of Lagrange Interpolation

- So, for the first term:

$$L_1 u_{j-1} = \frac{(x - x_j)(x - x_{j+1})}{(\Delta x)(2\Delta x)} u_{j-1} \quad (30)$$

- We expand the numerator (and collapse denominator):

$$L_1 u_{j-1} = \frac{(x^2 - xx_{j+1} - xx_j + x_j x_{j+1})}{(2\Delta x^2)} u_{j-1} \quad (31)$$

- Now we take the derivative w.r.t. x :

$$\frac{d(L_1)}{dx} u_{j-1} = \frac{(2x - x_{j+1} - x_j)}{(2\Delta x^2)} u_{j-1} \quad (32)$$

- When we set $x = x_{j-1}$ we get:

$$\frac{d(L_1)}{dx} u_{j-1} = \frac{(2x_{j-1} - x_{j+1} - x_j)}{(2\Delta x^2)} u_{j-1} = \frac{-3\Delta x}{(2\Delta x^2)} u_{j-1} = \frac{-3}{(2\Delta x)} u_{j-1} \quad (33)$$

FD Equations by the Method of Lagrange Interpolation

- So, for the second term of the summation:

$$L_2 u_j = - \frac{(x - x_{j-1})(x - x_{j+1})}{(\Delta x)(\Delta x)} u_j \quad (34)$$

- We expand the numerator (and collapse denominator):

$$L_2 u_j = - \frac{x^2 - xx_{j+1} - xx_{j-1} + x_{j+1}x_{j-1}}{(\Delta x)^2} u_j \quad (35)$$

- Now we take the derivative w.r.t. x :

$$\frac{dL_2}{dx} u_j = - \frac{2x - x_{j+1} - x_{j-1}}{(\Delta x)^2} u_j \quad (36)$$

- When we set $x = x_{j-1}$ we get (with some cleaning up):

$$\frac{dL_2}{dx} u_j = - \frac{2x_{j-1} - x_{j+1} - x_{j-1}}{(\Delta x)^2} u_j = - \frac{-2\Delta x}{(\Delta x)^2} u_j = \frac{4}{(2\Delta x)} u_j \quad (37)$$

FD Equations by the Method of Lagrange Interpolation

- Similarly, for Lagrange polynomials centered at different x points, we can find the following expressions when $x = x_{j-1}$:

$$\begin{aligned}\delta_{j-1}^m &= \frac{-3}{2\Delta x} \\ \delta_j^m &= \frac{2}{\Delta x} = \frac{4}{2\Delta x} \\ \delta_{j+1}^m &= \frac{-1}{2\Delta x}\end{aligned}$$

- Which, when put together yields the one-sided first derivative approximation:

$$\frac{du}{dx} = \left[\frac{-3}{2\Delta x} \right] u_j + \left[\frac{4}{2\Delta x} \right] u_{j+1} + \left[-\frac{1}{2\Delta x} \right] u_{j+2} = \frac{-3u_j + 4u_{j+1} - 1u_{j+2}}{2\Delta x}$$

- Recall: this was for $x = x_{j-1}$, so $u_{j-1} = u_j$.
- Try the final term for yourself!

FD Equations by the Method of Lagrange Interpolation

- The result is:

$$\frac{du}{dx} = \frac{-3u_j + 4u_{j+1} - u_{j+2}}{2\Delta x} + O(\Delta x^2)$$

- We have derived the equation for the one sided first derivative. Other equations can be generally derived in a similar fashion.
- Note: We had 3-points, and as such we approximated the function using a Lagrange polynomial of order 2.
- The error in this first derivative approximation is therefore second order ($O(\Delta x^2)$).

Summary: What have we learned?

- Finite differences are approximations to derivatives, using (finite) differences in values.
- Finite difference equations can be derived using the Taylor Series expansion formula or Lagrange Polynomial Interpolation.
- The Higher Order Taylor Series terms that are *ignored* in the finite difference approximation represent the error.
- As we make Δx smaller, a method with $H.O.T = O(\Delta x)$ will approach the true derivative slower than a method which is $H.O.T = O(\Delta x^2)$.
- Can derive finite difference approximations for (many) different derivatives.
- There are finite difference approximations of many different orders ($H.O.T = O(\Delta x^n)$, where n = order of derivative approximation)